A Linear-Algebra Guide to the Strange Life of High-Dimensional Data

Solutions to Questions 1–3 • SIMC 2026 Endeavour Challenge • 27 May 2026

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1 Markov matrices and the fate of a population

We are given two organism types whose fractional populations $p_1(t), p_2(t)$ evolve in discrete generations by $\boldsymbol{P}(t+1) = \mathbb{M}\,\boldsymbol{P}(t)$, where

$$\boldsymbol{P}(t) = \begin{pmatrix} p_1(t) \\ p_2(t) \end{pmatrix}, \qquad \mathbb{M} = \begin{pmatrix} 1-\varepsilon & 0 \\ \varepsilon & 1 \end{pmatrix}, \qquad \varepsilon \in (0,1). \tag{1}$$

Each step, a fraction ε of type 1 converts to type 2 and nothing returns. The two component equations are $p_1(t+1) = (1-\varepsilon)p_1(t)$ and $p_2(t+1) = \varepsilon\, p_1(t) + p_2(t)$.

1.1 (a) The total population is conserved

Let $S(t) = p_1(t) + p_2(t)$. Adding the two update equations, the ε terms cancel:

$$S(t+1) = (1-\varepsilon)p_1(t) + \varepsilon\, p_1(t) + p_2(t) = p_1(t) + p_2(t) = S(t). \tag{2}$$

So S never changes, and $S(t) = S(0)$ for all t: the total population is conserved.

1.2 (b) Eigenvalues and eigenvectors

Because $\mathbb{M}$ is lower triangular its eigenvalues are its diagonal entries, which we confirm from

$$\det(\mathbb{M} - \lambda\mathbf{I}) = \det\begin{pmatrix} 1-\varepsilon-\lambda & 0 \\ \varepsilon & 1-\lambda \end{pmatrix} = (1-\varepsilon-\lambda)(1-\lambda) = 0 \implies \lambda_1 = 1,\ \ \lambda_2 = 1-\varepsilon. \tag{3}$$

The eigenvectors come from solving $(\mathbb{M} - \lambda_i\mathbf{I})\boldsymbol{v}_i = \mathbf{0}$. For $\lambda_1 = 1$,

$$\mathbb{M} - \mathbf{I} = \begin{pmatrix} -\varepsilon & 0 \\ \varepsilon & 0 \end{pmatrix}, \qquad \begin{pmatrix} -\varepsilon & 0 \\ \varepsilon & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{0} \ \Rightarrow\ -\varepsilon x = 0 \ \Rightarrow\ x = 0 \text{ (y free)}, \tag{4}$$

so $\boldsymbol{v}_1 = (0,1)^\top$. For $\lambda_2 = 1-\varepsilon$,

$$\mathbb{M} - (1-\varepsilon)\mathbf{I} = \begin{pmatrix} 0 & 0 \\ \varepsilon & \varepsilon \end{pmatrix}, \qquad \varepsilon x + \varepsilon y = 0 \ \Rightarrow\ x + y = 0, \tag{5}$$

so $\boldsymbol{v}_2 = (1,-1)^\top$. The first eigenvector is the all-type-2 state that the dynamics never leaves; the second is the transient direction along which type 1 decays.

1.3 (c) Writing the initial state in the eigenbasis

With $\boldsymbol{P}(0) = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 = (c_2,\ c_1 - c_2)^\top$, matching components gives $c_2 = p_1(0)$ and $c_1 = p_2(0) + p_1(0) = S(0)$. So c_1 is exactly the conserved total from part (a), which is a useful consistency check rather than a coincidence. Taking the natural normalisation $p_1(0) + p_2(0) = 1$ with $p_1(0) = \frac{1}{2}$ gives the tidy decomposition $\boldsymbol{P}(0) = \boldsymbol{v}_1 + \frac{1}{2}\boldsymbol{v}_2$. We give the general coefficients first so that this normalisation assumption is stated explicitly.

1.4 (d) Closed-form evolution

The eigenbasis turns repeated matrix multiplication into ordinary powers. Applying $\mathbb{M}$ once,

$$\mathbb{M}\boldsymbol{P}(0) = \mathbb{M}(c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2) = c_1\mathbb{M}\boldsymbol{v}_1 + c_2\mathbb{M}\boldsymbol{v}_2 = c_1\lambda_1\boldsymbol{v}_1 + c_2\lambda_2\boldsymbol{v}_2, \tag{6}$$

since $\mathbb{M}\boldsymbol{v}_i = \lambda_i\boldsymbol{v}_i$. Each further step multiplies mode i by another λ_i, so $\mathbb{M}^t\boldsymbol{v}_i = \lambda_i^t\boldsymbol{v}_i$ and

$$\boldsymbol{P}(t) = \mathbb{M}^t\boldsymbol{P}(0) = c_1\lambda_1^t\boldsymbol{v}_1 + c_2\lambda_2^t\boldsymbol{v}_2 = c_1\,\boldsymbol{v}_1 + c_2\,(1-\varepsilon)^t\,\boldsymbol{v}_2, \tag{7}$$

using $\lambda_1 = 1$ and $\lambda_2 = 1-\varepsilon$. Each mode evolves on its own, scaled by its eigenvalue raised to the power t.

1.5 (e) Long-run behaviour

(i) The limit. Because $0 < \varepsilon < 1$ we have $0 < 1 - \varepsilon < 1$, so $(1-\varepsilon)^t \to 0$ as $t \to \infty$. The transient term in (d) therefore vanishes and

$$\boldsymbol{P}(t) \;\longrightarrow\; c_1\boldsymbol{v}_1 = \big(0,\; p_1(0) + p_2(0)\big)^\top, \tag{8}$$

the eigenvector with the larger eigenvalue $\lambda_1 = 1$; all mass ends up as type 2.

(ii) Why the structure forces this. The second column of $\mathbb{M}$ is $(0, 1)^\top$, so type 2 is absorbing: once an individual is type 2 it never leaves. Type 1, meanwhile, loses a fraction ε every step and is never replenished, so it drains away completely. The eigenvalue $\lambda_1 = 1$ carries the surviving steady state and $\lambda_2 = 1 - \varepsilon < 1$ governs how quickly the rest drains into it.

1.6 (f) Checking (d) and (e) numerically

We verified the closed form against a direct simulation that uses no analytical input: iterate $\boldsymbol{P}(t+1) = \mathbb{M}\boldsymbol{P}(t)$ for $T = 100$ generations at $\varepsilon = 0.1$, and compare with the formula from (d). The two agree to the last bit of double precision, with maximum discrepancy of order 10^{-16}, and the trajectory settles on $(0, 1)$ as predicted. The assertion is what makes this a test rather than a demonstration: if a later edit broke the formula, the cell would stop rather than quietly plot the wrong thing.

```
P = np.zeros((T+1, 2)); P[0] = p0 # track A: brute-force iteration
for s in range(T): P[s+1] = M @ P[s]
P_closed = c1*v1 + c2*((1-eps)**t)[:,None]*v2 # track B: closed form from (d)
assert np.max(np.abs(P - P_closed)) < 1e-10 # must agree to machine precision
```

Figure 1 shows the result. On the linear axis both populations move monotonically to their limits; on the semi-logarithmic axis the numeric $p_1(t)$ falls exactly along the line $\frac{1}{2}(1-\varepsilon)^t$, and a straight line there is the signature of geometric decay at rate $1 - \varepsilon$.

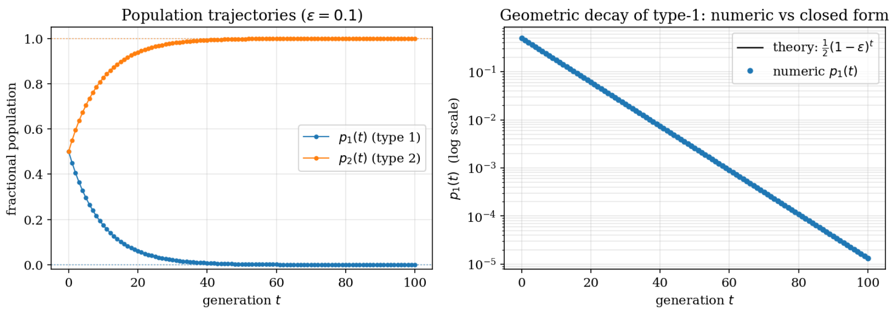

Figure 1: Constant $\varepsilon = 0.1$. Left: p_1, p_2 converge to $(0, 1)$. Right: the simulated $p_1(t)$ (markers) sits on the closed form $\frac{1}{2}(1-\varepsilon)^t$ (line) on a log axis, confirming both the formula of (d) and the geometric rate.

1.7 (g) A conversion rate that fades over time

Now let the rate decay, $\varepsilon(t) = \varepsilon_0 e^{-t/t_0}$, so the update matrix $\mathbb{M}(t)$ is different at every step. The key observation is that the eigenvectors do not depend on ε, so $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ are shared by every $\mathbb{M}(t)$: $\mathbb{M}(t)\boldsymbol{v}_1 = \boldsymbol{v}_1$ and $\mathbb{M}(t)\boldsymbol{v}_2 = (1 - \varepsilon(t))\boldsymbol{v}_2$. The state is the ordered product $\boldsymbol{P}(t) = \mathbb{M}(t-1)\cdots\mathbb{M}(0)\,\boldsymbol{P}(0)$. Acting on $\boldsymbol{P}(0) = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2$, the persistent mode is multiplied by 1 at every step, while the transient mode picks up one factor $1 - \varepsilon(s)$ at step s; after t steps these factors have accumulated into a product:

$$\boldsymbol{P}(t) = c_1\boldsymbol{v}_1 + c_2\Big[\textstyle\prod_{s=0}^{t-1}\big(1 - \varepsilon_0 e^{-s/t_0}\big)\Big]\boldsymbol{v}_2. \tag{9}$$

So the geometric power $(1-\varepsilon)^t$ of part (d) becomes a *product* $\prod_s(1 - \varepsilon(s))$; the power $(1 - \varepsilon(t))^t$ would be correct only if the matrix were fixed.

This product behaves very differently from the constant-rate case, which is the interesting part. With ε fixed, every factor is the same number below 1, so the product shrinks all the way to 0 and type 1 disappears. With a *fading* rate the factors $1 - \varepsilon(s)$ climb back towards 1 as $\varepsilon(s) \to 0$, and they reach 1 fast enough that the running product stops changing and settles at a strictly positive value Π_∞ instead of 0 – once we are multiplying by numbers indistinguishable from 1, the product barely moves. The conversion has effectively switched off, freezing a residue $c_2\Pi_\infty$ in the $\boldsymbol{v}_2$ direction, so the population settles at $\boldsymbol{P}(\infty) = c_1\boldsymbol{v}_1 + c_2\Pi_\infty\boldsymbol{v}_2$

and type 1 is *never fully depleted*. A fading conversion rate therefore traps a small permanent population of type 1, the qualitative correction to parts (d)–(e).

1.8 (h) Checking the fading rate

We validated the product formula exactly as in (f), iterating with a freshly built $\mathbb{M}(s)$ at each step ($\varepsilon_0 = 0.5$, $t_0 = 10$) and comparing against the closed form; agreement is again at the 10^{-16} level. Figure 2 overlays the fading-rate trajectory on the constant-rate case. The contrast is the whole point: with constant ε, p_1 heads for zero in a straight log-line, whereas with $\varepsilon(t)$ it bends and locks onto a positive plateau once the rate has effectively vanished. The measured residue matches the small-rate estimate $\frac{1}{2}e^{-\varepsilon_0 t_0}$ to the expected accuracy.

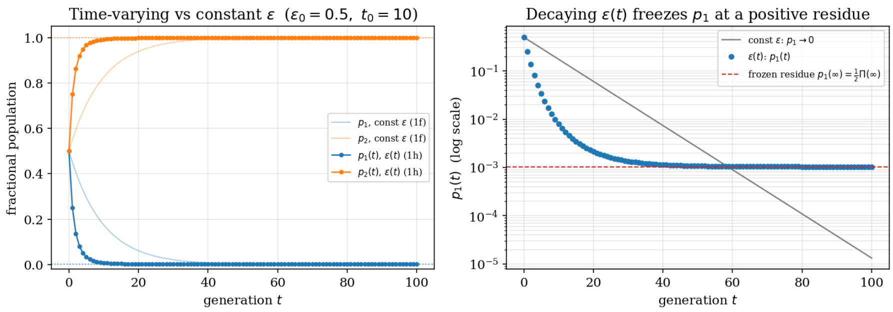

Figure 2: Fading rate $\varepsilon(t) = \varepsilon_0 e^{-t/t_0}$ (solid) against constant ε (faded). Right panel, log axis: the constant case drains to zero, while the fading case freezes at the positive residue $c_2\Pi_\infty$ (dashed).

2 The power method and the singular value decomposition

Here $\mathbf{X}$ is a symmetric $n \times n$ matrix with eigenvalues $|\lambda_1| > |\lambda_2| \geq \cdots \geq |\lambda_n|$ and an orthonormal eigenbasis $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$. The power method starts from a unit vector and repeats $\boldsymbol{x}_{k+1} = \mathbf{X}\boldsymbol{x}_k / \|\mathbf{X}\boldsymbol{x}_k\|$.

2.1 (a) Why the iteration finds the dominant eigenvector

Claim. If $\boldsymbol{x}_0$ has a nonzero component along $\boldsymbol{v}_1$, then $\boldsymbol{x}_k \longrightarrow \pm\boldsymbol{v}_1$.

Normalisation only rescales, so $\boldsymbol{x}_k$ points the same way as $\boldsymbol{y}_k = \mathbf{X}^k \boldsymbol{x}_0$. Expand $\boldsymbol{x}_0 = \sum_i a_i \boldsymbol{v}_i$ with $a_1 \neq 0$. Since $\mathbf{X}^k \boldsymbol{v}_i = \lambda_i^k \boldsymbol{v}_i$,

$$\boldsymbol{y}_k = a_1 \lambda_1^k \boldsymbol{v}_1 + \sum_{i \geq 2} a_i \lambda_i^k \boldsymbol{v}_i. \tag{10}$$

Measuring the off-axis part relative to the leading term, and using orthonormality,

$$\rho_k = \frac{\left\|\sum_{i\geq 2} a_i \lambda_i^k \boldsymbol{v}_i\right\|}{|a_1|\,|\lambda_1|^k} = \frac{1}{|a_1|}\sqrt{\sum_{i\geq 2} a_i^2 \left(\frac{\lambda_i}{\lambda_1}\right)^{2k}}. \tag{11}$$

Every ratio $|\lambda_i/\lambda_1| < 1$, so each term vanishes and $\rho_k \longrightarrow 0$: the iterate aligns with $\boldsymbol{v}_1$, and after normalising, $\boldsymbol{x}_k \longrightarrow \pm\boldsymbol{v}_1$, with the sign tracking $\operatorname{sign}(a_1\lambda_1^k)$.

2.2 (b) Rate of convergence and the spectral gap

The same expression bounds how fast this happens. Writing $\omega_i = a_i^2 \lambda_i^{2k} \geq 0$,

$$\frac{\rho_{k+1}}{\rho_k} = \frac{1}{|\lambda_1|}\sqrt{\frac{\sum_{i\geq 2}\omega_i \lambda_i^2}{\sum_{i\geq 2}\omega_i}} \leq \frac{1}{|\lambda_1|}\sqrt{\lambda_2^2} = \left|\frac{\lambda_2}{\lambda_1}\right| < 1, \tag{12}$$

because the fraction under the root is a weighted average of the λ_i^2 ($i \geq 2$), which cannot exceed the largest, λ_2^2. Convergence is therefore geometric at rate $|\lambda_2/\lambda_1|$. The quantity that controls everything is the spectral gap between the two largest eigenvalues: a wide gap means fast, stable convergence, a narrow one means slow and fragile.

2.3 (c) When the gap closes

Take $\mathbf{X} = \text{diag}(1, -1)$, so $|\lambda_1| = |\lambda_2| = 1$ and there is no gap. For a generic start (a, b) one finds $\mathbf{X}^k(a, b)^\top = (a, (-1)^k b)^\top$. The norm is preserved, so after normalisation the iterate flips forever between (a, b) and $(a, -b)$ and never settles. This is not merely slow convergence but a structural failure. The power method works by amplifying each eigencomponent by its own factor λ_i^k, and the top one wins only because it is strictly largest. When two leading eigenvalues share a magnitude, their relative weights never change and there is no mechanism to select one: the iterate keeps whatever mixture it began with, oscillating when the signs differ or, if $\lambda_1 = \lambda_2$, drifting to an arbitrary vector of the shared eigenspace. The near-degenerate case is worse in practice than in theory. The iteration count scales like $1/(1 - |\lambda_2/\lambda_1|)$ and blows up, while the useful per-step separation between $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ shrinks to the level of the 10^{-16} rounding error that every floating-point step injects, at which point the computed ordering of the two eigenvalues becomes ambiguous.

2.4 (d) The matrices $\mathbf{X}^\top\mathbf{X}$ and $\mathbf{X}\mathbf{X}^\top$

Let $\mathbf{X}$ now be a general $m \times n$ matrix. Both products are symmetric, since $(\mathbf{X}^\top\mathbf{X})^\top = \mathbf{X}^\top\mathbf{X}$ and likewise for $\mathbf{X}\mathbf{X}^\top$, and both are positive semi-definite, because for any $\boldsymbol{w}$,

$$\boldsymbol{w}^\top(\mathbf{X}^\top\mathbf{X})\boldsymbol{w} = \|\mathbf{X}\boldsymbol{w}\|^2 \geq 0, \qquad \boldsymbol{w}^\top(\mathbf{X}\mathbf{X}^\top)\boldsymbol{w} = \|\mathbf{X}^\top\boldsymbol{w}\|^2 \geq 0. \tag{13}$$

Being symmetric and positive semi-definite, each has real non-negative eigenvalues, so $\sigma_i = \sqrt{\lambda_i}$ is well defined. They also share their nonzero eigenvalues. If $\mathbf{X}^\top\mathbf{X}\boldsymbol{v} = \lambda\boldsymbol{v}$ with $\lambda \neq 0$, then left-multiplying by $\mathbf{X}$,

$$(\mathbf{X}\mathbf{X}^\top)(\mathbf{X}\boldsymbol{v}) = \lambda(\mathbf{X}\boldsymbol{v}), \tag{14}$$

and $\mathbf{X}\boldsymbol{v} \neq \mathbf{0}$ (otherwise $\lambda\boldsymbol{v} = \mathbf{X}^\top\mathbf{X}\boldsymbol{v} = \mathbf{0}$ would force $\lambda = 0$). So $\mathbf{X}\boldsymbol{v}$ is an eigenvector of $\mathbf{X}\mathbf{X}^\top$ with the same eigenvalue, and the symmetric argument gives the converse. The shared nonzero values are the σ_i^2. This is the fact Question 3 leans on to connect its covariance and Gram analyses.

2.5 (e) A power-method SVD, and a worked example

To find the leading singular triplet of an $N \times M$ matrix $\mathbf{X}$, run the power method on whichever of $\mathbf{X}^\top\mathbf{X}$ $(M \times M)$ or $\mathbf{X}\mathbf{X}^\top$ $(N \times N)$ is smaller, since both square to size $\min(N, M)$ and share the spectrum of interest. Recover the top eigenvector $\boldsymbol{v}_1$, set $\sigma_1 = \sqrt{\boldsymbol{v}_1^\top(\mathbf{X}^\top\mathbf{X})\boldsymbol{v}_1}$ from the Rayleigh quotient, and obtain the partner vector by a single multiplication, $\boldsymbol{u}_1 = \mathbf{X}\boldsymbol{v}_1/\sigma_1$.

Concretely, take $\mathbf{X} = \left(\begin{smallmatrix}1&0\\0&1\\1&1\end{smallmatrix}\right)$ with $N = 3, M = 2$. The smaller matrix is $\mathbf{X}^\top\mathbf{X} = \left(\begin{smallmatrix}2&1\\1&2\end{smallmatrix}\right)$, with eigenvalues 1 and 3. The dominant one gives $\lambda_1 = 3$, $\sigma_1 = \sqrt{3}$, and $\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(1, 1)^\top$. Then $\boldsymbol{u}_1 = \mathbf{X}\boldsymbol{v}_1/\sigma_1 = \frac{1}{\sqrt{6}}(1, 1, 2)^\top$. Both defining relations hold: $\mathbf{X}\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(1, 1, 2)^\top = \sigma_1\boldsymbol{u}_1$, and $\mathbf{X}^\top\boldsymbol{u}_1 = \frac{3}{\sqrt{6}}(1, 1)^\top = \sigma_1\boldsymbol{v}_1$, confirming the triplet by hand.

3 Cleaning, principal components, and clustering of spectra

The third question hands us `mystery_matrix1.npy`, a collection of infrared absorbance spectra. We treat it as a design matrix $\mathbf{X} \in \mathbb{R}^{N \times M}$ with rows indexing samples (vials) and columns indexing features (frequency channels), so $\mathbf{X}_{ij}$ is the absorbance of vial i at frequency j.

3.1 (a) Orienting and cleaning the matrix

The file loads as a 3401×1000 array. Two things are worth saying before any analysis. First, the array is stored transposed relative to the row-as-sample convention, so we transpose once so that rows are samples. Second, the prompt quotes $M = 3041$ features whereas the file has 3401 columns, a digit transposition; we use the array's true dimension and key off the axis of length 1000 rather than hard-coding either number.

An empty vial records only background noise, which is small and flat, so it has a small Euclidean norm $\|\boldsymbol{x}_i\| = (\sum_j \mathbf{X}_{ij}^2)^{1/2}$. Sorting the 1000 norms makes the structure plain: 40 vials cluster near $\|\boldsymbol{x}\| \approx 1.6$, then there is an empty gap, and the bulk begins near 4.3. We place the threshold in that gap. Crucially the count of flagged vials does not move for any cutoff between 2.0 and 4.0, which is the evidence that the cut is principled rather than tuned. This removes exactly $T = 40$ empties (rows 960–999). As an independent check, the removed spectra are 50.2% exact zeros, consistent with background noise clipped at zero (which is exactly zero with probability about a half), against 24.5% for the genuine spectra.

A second defect is hiding in the upper tail, one the prompt does not mention. Thirty-six vials sit near $\|\boldsymbol{x}\| \approx 8.9$ (rows 924–959), separated from the bulk by a gap roughly five times wider than any gap inside it. We keep a vial only if its norm lies between the two cuts, implemented as a Boolean mask so the operation is non-destructive and index-preserving, which leaves $\widetilde{N} = 924$ spectra. We were initially reluctant to discard a second group on a hunch, so we tested it: in the top-four principal-component space the 36 form one tight, isolated cluster, and removing them lowers the total cluster count by exactly one while leaving every other group untouched (Figure 3, right). The decision is therefore clean and, because it is a mask, entirely reversible.

Figure 3: Cleaning. Left: the row-norm distribution, with 40 empty vials below a clean gap (threshold dashed). Right: the 36 high-norm vials (red) form a single isolated cluster in the leading PC plane; removing them deletes exactly one cluster.

3.2 (b) Feature variance and the covariance matrix

We mean-centre each feature, $\widetilde{\mathbf{X}}_{ij} = \mathbf{X}_{ij} - \bar{\mathbf{X}}_j$, which removes the shared average spectrum so the analysis describes how samples differ rather than what they have in common. The covariance matrix is $\mathbf{C} = \frac{1}{\widetilde{N}}\widetilde{\mathbf{X}}^\top\widetilde{\mathbf{X}} \in \mathbb{R}^{M\times M}$. Its diagonal entries are exactly the feature variances, $\mathbf{C}_{jj} = \frac{1}{\widetilde{N}}\sum_k \widetilde{\mathbf{X}}_{kj}^2 = \mathrm{Var}_j$, since the columns now have zero mean, and it is symmetric because $(\widetilde{\mathbf{X}}^\top\widetilde{\mathbf{X}})^\top = \widetilde{\mathbf{X}}^\top\widetilde{\mathbf{X}}$. The trace is the total variance, $\operatorname{tr}\mathbf{C} = \sum_j \mathrm{Var}_j$, and it is invariant under any orthonormal change of basis, $\operatorname{tr}(\mathbf{Q}^\top\mathbf{C}\mathbf{Q}) = \operatorname{tr}(\mathbf{C}\mathbf{Q}\mathbf{Q}^\top) = \operatorname{tr}\mathbf{C}$, so it also equals $\sum_i \lambda_i$. Numerically $\operatorname{tr}\mathbf{C} \approx 7.90$, and we use the identity $\sum_i \lambda_i = \operatorname{tr}\mathbf{C}$ later as a check on the eigendecomposition.

The single most variable feature is $j^\star = 910$, with $\mathbf{C}_{j^\star j^\star} \approx 0.0199$. More telling is the shape of the variance profile (Figure 4): it is not spread evenly across the 3401 channels but concentrates in a few narrow bands, with broad flat stretches of baseline in between. The profile itself looks like a spectrum. Those bands are the frequencies where the dominant molecules absorb and where vials genuinely differ; the flat regions carry almost no discriminating information. The data is, in other words, highly redundant, which is precisely the structure principal component analysis exploits next.

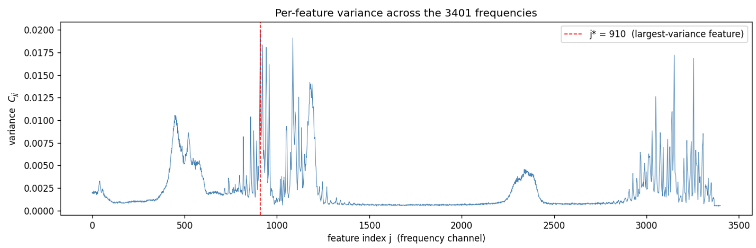

Figure 4: Per-feature variance $\mathbf{C}_{jj}$ across the 3401 frequencies, with the maximum at $j^\star = 910$ (dashed). Variance lives in a handful of molecular bands, not the baseline.

3.3 (c) The eigenspectrum

For a unit direction $\boldsymbol{w}$ the variance of the projected data is $\boldsymbol{w}^\top\mathbf{C}\boldsymbol{w}$, and for an eigenvector this equals the eigenvalue; since $\boldsymbol{w}^\top\mathbf{C}\boldsymbol{w} = \frac{1}{\widetilde{N}}\|\widetilde{\mathbf{X}}\boldsymbol{w}\|^2 \geq 0$, every $\lambda_i \geq 0$. By the spectral theorem $\mathbf{C}$ has an orthonormal eigenbasis, so we diagonalise it with `numpy.linalg.eigh`, the solver for symmetric matrices, which returns real eigenvalues and orthonormal eigenvectors and is both faster and more stable than the general `eig`. We sort the eigenvalues in decreasing order and clip rounding-level negatives to zero.

```
C = (Xc.T @ Xc) / Nt # covariance (M x M); Xc is mean-centred X
```

```
w, V = np.linalg.eigh(C) # symmetric solver: real, orthonormal
order = np.argsort(w)[::-1]
eigvals = np.clip(w[order], 0, None); eigvecs = V[:, order]
kstar = int(np.argmax(eigvals[:30] / eigvals[1:31])) + 1 # elbow = largest consecutive ratio
```

The leading eigenvalues are tabulated below. The ratio of the two largest is $\lambda_1/\lambda_2 \approx 3.53$, and λ_1 alone accounts for about 31% of the total variance, so one direction clearly dominates the variation. A large ratio of this kind is exactly the wide spectral gap that, by Question 2, would make the power method converge quickly here, and it signals that the data has strong low-dimensional structure.

k	1	2	3	4	5	6	7	8	9
λ_k	2.492	0.705	0.590	0.254	0.123	0.105	0.078	0.039	0.0093

Table 1: Leading eigenvalues. The drop $\lambda_8/\lambda_9 \approx 4.1$ is the largest in the spectrum and marks the elbow.

To find the number of meaningful components we look for the elbow, the largest multiplicative drop between consecutive eigenvalues. We use a ratio rather than a difference because the boundary between signal and noise is multiplicative on the logarithmic scale of Figure 5. The largest drop is $\lambda_8/\lambda_9 \approx 4.1$, giving $k^\star = 8$. On the log plot the first eight eigenvalues fall steeply and the rest lie on a near-flat noise floor of around 900 tiny, slowly-decreasing values. Eight signal components is satisfyingly close to the roughly ten dominant molecules referred to in part (h): the spectra are mixtures of a small number of chemical components, so only a handful of independent directions carry real signal.

Intrinsic dimension

The nominally 3401-dimensional data has only $k^\star = 8$ signal directions. The elbow is set by the largest multiplicative drop, $\lambda_8/\lambda_9 \approx 4.1$; everything beyond it lies on a flat noise floor. Eight signal dimensions echoes the ~ 10 dominant molecules of part (h).

It is tempting to instead keep "95% of the variance", but here that rule is misleading. The eight signal components explain only about 55% of the variance, and reaching 95% takes 667 components (Table 2), because the noise floor, though negligible term by term, is spread over hundreds of dimensions that sum to a large share. A 95% cut would drag in some 660 noise directions and badly overstate the dimensionality. The elbow is the honest measure.

cumulative variance	50%	80%	90%	95%	99%
components k	4	309	516	667	850

Table 2: Components needed to reach each variance threshold. Contrast the eight signal components (about 55%) with the 667 required for 95%.

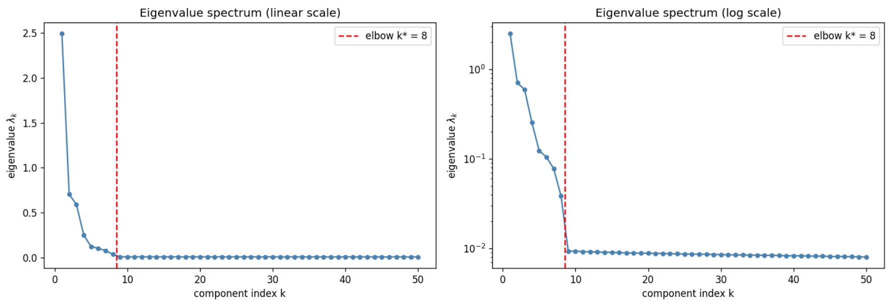

Figure 5: Eigenvalue spectrum on linear (left) and logarithmic (right) axes. The log plot shows a sharp elbow at $k^\star = 8$ followed by a flat noise floor.

A threshold-free reading agrees with the elbow. The explained-variance ratio of component k and the *participation ratio* (a continuous effective dimension) are

$$r_k = \frac{\lambda_k}{\sum_j \lambda_j}, \qquad d_{\text{eff}} = \frac{\left(\sum_i \lambda_i\right)^2}{\sum_i \lambda_i^2}. \tag{15}$$

The participation ratio needs no cutoff: it is large when many eigenvalues share the variance and small when a few dominate. Here $d_{\text{eff}} \approx 8.7$, in close agreement with the elbow at $k^\star = 8$ and the ~ 10 dominant molecules — three independent criteria (the multiplicative drop, the participation ratio, and the chemistry) converging on the same handful of dimensions.

3.4 (d) Reading the leading eigenvector

The leading eigenvector $\boldsymbol{v}_1 \in \mathbb{R}^M$ assigns a loading to each frequency, and the scores $z_i = \widetilde{\mathbf{x}}_i \cdot \boldsymbol{v}_1$ are the one-dimensional summaries of greatest spread. Eigenvectors are defined only up to sign, so we fix the largest loading positive. The loadings are not spread evenly: they concentrate on two narrow bands, near $j \approx 900$–960 and 3150–3260, and about 27% of the entries are negative. So $\boldsymbol{v}_1$ is a contrast direction, measuring the balance of absorption between bands, not overall brightness. Using cosine similarity, which is scale-free, $\boldsymbol{v}_1$ matches a mean-centred spectrum at about 0.92 but the raw spectrum at only 0.67 and the mean spectrum at 0.03. Living in the centred space by construction, it is nearly orthogonal to the shared mean and behaves like a difference spectrum (Figure 6): it captures how the most distinctive vial departs from the average, which is the chemical contrast that separates the mixtures most strongly. The echo of Question 1 is hard to miss, where the dominant eigenvector again carried the structure that survived.

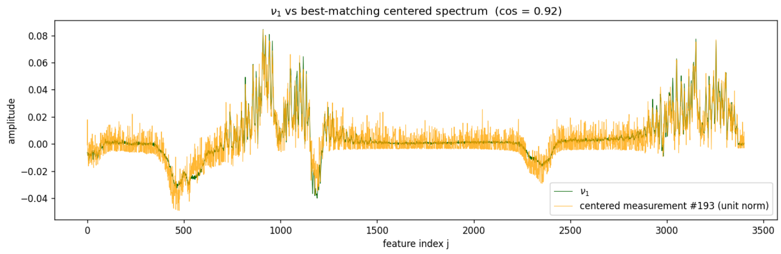

Figure 6: The leading eigenvector $\boldsymbol{v}_1$ (green) overlaid on the best-matching centred spectrum (cosine ≈ 0.92). It reads as a difference spectrum supported on two molecular bands.

3.5 (e) Clusters in covariance space

Projecting each vial onto the top four eigenvectors gives compressed coordinates $\mathbf{Z} = \widetilde{\mathbf{X}}\mathbf{V}_4 \in \mathbb{R}^{\widetilde{N}\times 4}$, four scores per vial. The six pairwise scatter plots (Figure 7) show many tight, well-separated blobs. To count them without reaching for a library we use leader clustering, starting a new cluster whenever a point is farther than a radius r from every existing centre.

```
def count_clusters(P, r): # numpy-only, no sklearn
    centres = [P[0]]
    for p in P[1:]:
        if np.sqrt(((np.array(centres) - p)**2).sum(1)).min() > r:
            centres.append(p)
    return len(centres)
```

The count is 25, and it is stable for any radius between about six and twelve times the median nearest-neighbour spacing, so it reflects the data rather than the threshold. Each blob is a set of vials with nearly identical spectra, that is, one chemical mixture, so the 924 cleaned vials are about 25 recipes, comfortably under the "fewer than 100" mentioned in part (h). Although the eigenvectors live in feature space, projecting samples onto them exposes sample-space structure, because similar spectra receive similar scores.

3.6 (f) The same question through the Gram matrix

The Gram matrix $\mathbf{G} = \frac{1}{\widetilde{N}}\widetilde{\mathbf{X}}\widetilde{\mathbf{X}}^\top \in \mathbb{R}^{\widetilde{N}\times\widetilde{N}}$ lives in sample space, with $\mathbf{G}_{ij}$ the similarity between vials i and j. Because $\widetilde{N} = 924 \ll M = 3401$, the Gram matrix is only $\widetilde{N} \times \widetilde{N}$ rather than $M \times M$, so diagonalising it costs $(M/\widetilde{N})^2 \approx 13.5$ times less than the covariance route — exactly the "use the smaller matrix" rule of Question 2(e). By the result of Question 2(d) it shares the nonzero eigenvalues of $\mathbf{C}$, which we confirm numerically: the top eight Gram eigenvalues match the top eight covariance eigenvalues to within 10^{-6}. A Gram eigenvector's components are themselves the sample coordinates, with no projection of $\widetilde{\mathbf{X}}$ needed; scaling by $\sqrt{\widetilde{N}\lambda}$ matches the units of part (e). The six Gram scatter plots reproduce those of the covariance analysis up to per-axis sign flips, and the same leader-clustering rule again returns 25 clusters (the two sets

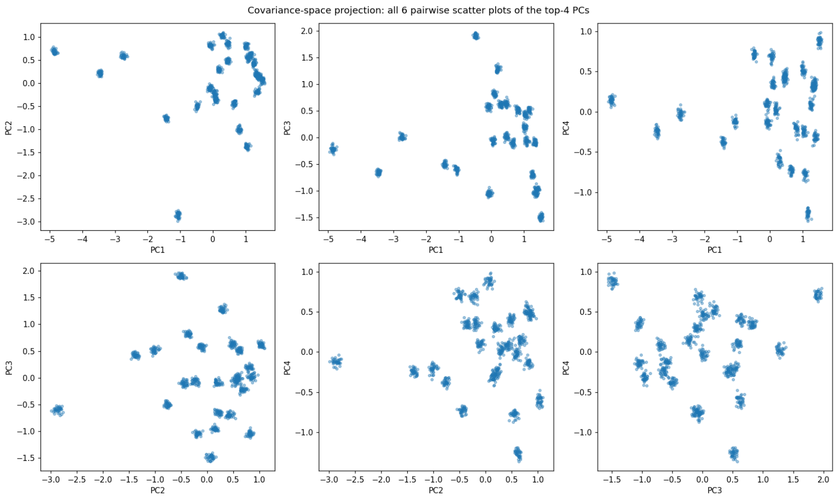

Figure 7: The six pairwise scatter plots of the top-four covariance scores. Each point is a vial; the data resolves into about 25 clusters.

of scores are overlaid explicitly in Figure 8 of part g). Whether we view the data through its features or through sample-to-sample similarity, the structure is the same.

3.7 (g) Are the two clusterings really the same?

They are, and necessarily so. If $\boldsymbol{\nu}$ is an eigenvector of $\mathbf{C} = \frac{1}{N}\widetilde{\mathbf{X}}^{\top}\widetilde{\mathbf{X}}$ with eigenvalue λ, then $\widetilde{\mathbf{X}}\boldsymbol{\nu}$ is an eigenvector of $\mathbf{G} = \frac{1}{\widetilde{N}}\widetilde{\mathbf{X}}\widetilde{\mathbf{X}}^{\top}$ with the same λ, so the two matrices share their nonzero spectrum. The cleaner statement uses the singular value decomposition. Writing $\widetilde{\mathbf{X}} = \mathbf{U}\Sigma\mathbf{V}^{\top}$, the covariance and Gram score matrices are

$$\mathbf{Z} = \widetilde{\mathbf{X}}\mathbf{V} = \mathbf{U}\Sigma, \qquad \mathbf{Z}_{\mathrm{g}} = \mathbf{U}\sqrt{\widetilde{N}\lambda} = \mathbf{U}\Sigma, \tag{16}$$

the *same* matrix $\mathbf{U}\Sigma$, identical up to the arbitrary sign of each eigenvector. We checked this directly: the 4×4 correlation matrix between the two score sets is the identity in absolute value, 1 on the diagonal and 0 off it, and the coordinates agree in magnitude to about 10^{-14}, with the first and fourth axes happening to come out sign-flipped (Figure 8). Because the two point clouds are the same up to reflecting individual axes, the clusters and their count must coincide, which is why both routes give 25.

The two views answer different-sounding questions that turn out to be one question. Covariance space asks which combination of frequencies a vial expresses; Gram space asks which other vials it resembles. Both place vials with nearly identical spectra at the same point, so proximity in either coordinate system means the same chemical mixture.

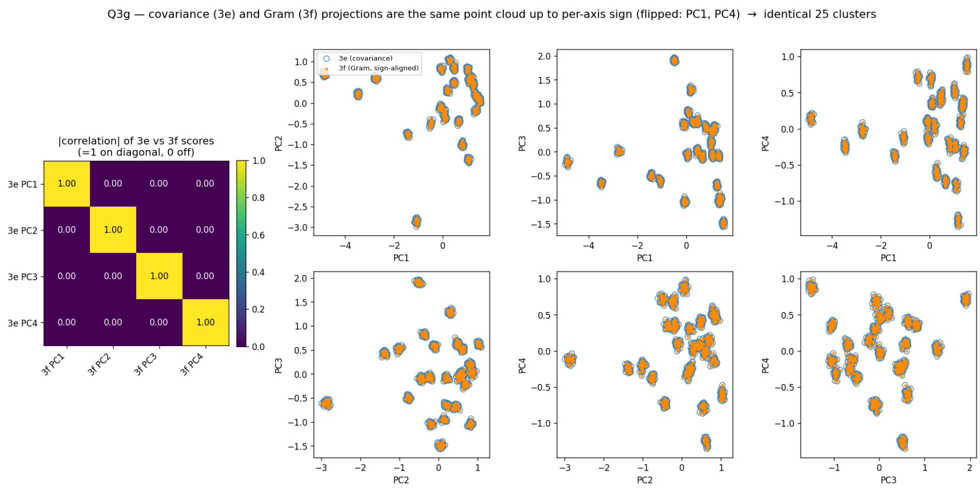

Figure 8: Covariance (3e) against Gram (3f). Left: the absolute correlation between the two score sets, the identity. Right: the six PC-pair scatters with covariance scores (blue circles) and sign-aligned Gram scores (orange dots) overlaid; they coincide point for point, so the two methods return the identical 25 clusters.

3.8 (h) The forgetful taste tester

The brief: the 1000 vials hold fewer than 100 unique fermented-drink samples (one per vial), fingerprinted by ten dominant molecules whose absorption bands the spectroscopy targets.

(i) How many unique samples were there? Our analysis answers this directly: about **25**. After discarding the 40 empty vials and the 36 corrupted high-norm vials, the 924 genuine spectra fall into 25 tight clusters, and each cluster is a set of nearly identical spectra, that is, one recipe. We trust the figure because it is stable rather than fitted: it holds across clustering radii spanning a factor of two, and the same 25 emerges from two independent constructions, the covariance matrix and the Gram matrix (part g). It also sits well under the stated bound of one hundred. The ten dominant molecules surface as the eight signal dimensions of the eigenspectrum: each spectrum is a non-negative mixture of about ten absorption signatures, so the mean-centred data needs only that many independent directions, and PCA recovers them as the $k^\star = 8$ components standing above the noise floor. (The 36 high-norm vials are a measurement fault, not a 26th recipe; part (a) shows they form a single corrupted group.)

The 25 groups are also near-uniform in size — 24 of them hold 37 replicate vials and one holds 36, so $24 \times 37 + 36 = 924$ exactly (Figure 9). A near-constant replicate count is the fingerprint of a designed taste test that measured a fixed menu of recipes several dozen times each, which is exactly what we would expect once the lost labels are restored.

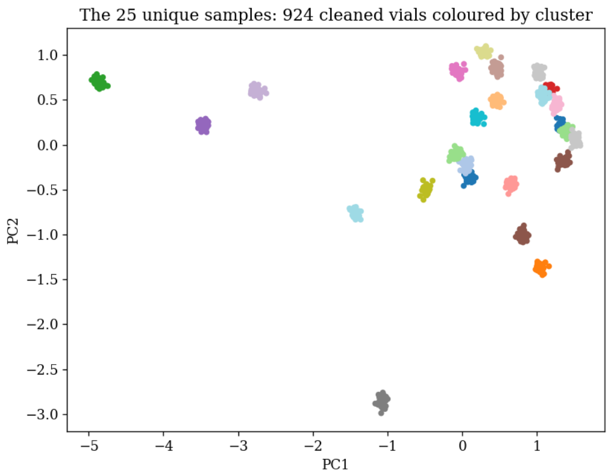

Figure 9: The 924 cleaned vials, coloured by cluster, resolve into 25 well-separated groups in the leading covariance-score plane. The groups are near-uniform in size ($24 \times 37 + 36 = 924$), the signature of a designed experiment with 25 unique samples.

Answer to Question 3(h)

There were about 25 **unique food samples**. The 924 genuine spectra form 25 tight clusters of ~37 replicate vials each; the count is stable across a factor-of-two range of clustering radii and is reproduced independently by the covariance and Gram routes (part g), which by the Question 2(d) duality is a genuine cross-check rather than a coincidence.

(ii) Which earlier parts made this possible? Almost all of them, which is the real lesson of the question. The cleaning in (a) was a precondition, since the empties and especially the outliers would otherwise have invented spurious clusters and skewed the variance. The mean-centring and covariance of (b) and the eigenspectrum of (c) compressed 3401 noisy features down to the eight that carry signal, without which the clusters are invisible to the eye. The projection and clustering of (e) produced the count, the Gram route of (f) reproduced it, and the comparison in (g) turned that agreement into a genuine cross-check rather than a coincidence, resting on the $\mathbf{X}^\top\mathbf{X}$ / $\mathbf{X}\mathbf{X}^\top$ duality proved in Question 2(d). Even Question 1 supplies the guiding idea, that the leading eigenvectors of the right matrix carry the structure a system settles into. Our estimate of 25 is therefore not one measurement but the consistent reading of the whole pipeline.

4 Validation, robustness, and limitations

We tried to make every headline number answerable by more than one route. In Question 1 each closed form is checked against a direct iteration to the limit of double precision. In Question 3 the trace identity $\operatorname{tr}\mathbf{C} = \sum_i \lambda_i \approx 7.90$ confirms the eigendecomposition, the Gram and covariance spectra agree to 10^{-6}, and the two clusterings agree exactly, which is the empirical face of the Question 2(d) theorem. The hand-worked SVD example satisfies both defining relations.

The results are stable where it matters. The empty-vial count does not change for any threshold in $[2.0, 4.0]$, and the cluster count holds for clustering radii spanning a factor of two, so neither headline depends on a tuned parameter. The elbow at $k^\star = 8$ rests on a fourfold multiplicative drop, far clear of the noise floor's near-unit ratios.

The compression is principled rather than ad hoc. By the Eckart–Young theorem, projecting onto the top k eigenvectors is the *best* rank-k approximation of $\widetilde{\mathbf{X}}$ in the least-squares sense — the same singular value decomposition of Question 2, now read as an approximation rather than a factorisation, with squared error equal to the sum of the discarded eigenvalues $\sum_{i>k} \lambda_i$. Keeping $k^\star = 8$ replaces each 3401-number spectrum by 8 scores, a 425-fold reduction, while retaining the $\approx$ 55% of the variance that genuinely separates the mixtures; the discarded $\approx$ 45% is the noise floor, spread so thinly — about 0.05% per direction over the remaining ~900 dimensions — that no single dropped component carries recoverable structure. This is why a handful of directions suffices to recover the 25 recipes from a nominally 3401-dimensional measurement.

We are also clear about the limits. Leader clustering is a greedy method, and although the count is stable we do not claim it is provably optimal; a pathological radius could merge or split the loosest blobs. We identify the number of mixtures, not their chemical identities, which would need the actual absorption bands. And we resolve the 3401-versus-3041 discrepancy in favour of the file while assuming that was the setters' intent.

5 Conclusion

The three questions rhyme. A draining population, a convergent iteration, and a redundant spectral dataset are all governed by the eigenvalues of an appropriate matrix, and in each case the leading eigenvalue and the gap below it decide the outcome: type 2 wins unless a fading rate freezes the population short; the power method converges at a speed set by the spectral gap; and the spectra, nominally in thousands of dimensions, really occupy about eight and sort into roughly 25 mixtures that the covariance and Gram matrices agree upon. Given another week we would identify the molecular band behind each principal component and turn the 25 anonymous clusters into 25 named recipes.

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Statement on the use of AI

We used Claude Opus 4.7 (via Claude Code) for bounded, non-decisional support only: drafting code comments, wording figure captions and LaTeX, generating plotting code to our own specification, and checking our algebra for slips. The modelling, the proofs, every reported number, and all interpretation are our own, and we reviewed, edited, and re-ran each suggestion before using it. The full prompt-and-response record is in the companion file `SIMC10_PSBB-KKN_ailog.pdf`.

Reproducibility. Every figure and number above regenerates top-to-bottom from the single notebook `main.ipynb` (`numpy` and `matplotlib` only); the pipeline uses exact eigendecomposition and order-based clustering with no random seed, so $\widetilde{N} = 924$, the eigenspectrum, the elbow $k^\star = 8$, and the 25 clusters reproduce on every run.